



A Smart Scholarship Management System: All-in-One Solution

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Abstract: This paper proposes a smart Scholarship Management System (SMS) specifically designed for organizations providing scholarships for students in Lebanon. The SMS system aims to enhance efficiency and introduce new concepts to digitalized scholarship management. It will utilize web programming to provide accessibility and usability across devices. By integrating sentiment analysis, calculating the top 20 students, budget tracking, detailed student and donors' profiles, applying dashboard, generating reports, smart donor matching, receipt scanning, and role-based access, the expected outcome is a robust platform that automates the administrative processes and eases the student's experience in applying for scholarships while maintaining transparency and accountability. By implementing the advanced features of this system, it will fill significant gaps where such systems and tools are absent in Lebanon, and combined, absent worldwide. Overall, enhancing access to education for underserved communities by enabling more students to achieve their educational and career goals is crucial in this system's broader context. This paper will discuss the methodology and results and will highlight the significance of the project.

Keywords: Scholarship Management, Donor Management, Budget Management, Sentiment Analysis, Student Information System, Transparency

Introduction

In Lebanon, access to educational scholarships plays a crucial role in empowering students to pursue higher education and build successful careers while the country is consistently facing crises and surprising events. NGO's and other organizations work hard to provide scholarships. However, many still rely on outdated processes or fragmented digital tools. This reliance may lead to errors or inefficiencies and requires more effort and time. This system will be built within the scope of software and web application development, it is scalable and will seamlessly integrate a robust backend, it will provide a solution with great data management practices tailored to match the needs. Specific objectives include building a web-based app with data-driven decisions and calculations, implementing sentiment analysis to gain insights into scholarship essays submitted by students, creating a secure personalized admin, staff, and user's dashboards to ensure controlled access and data integrity, in addition to calculating the top 20 students from the applicants based on several factors, developing reporting options, automated budget tracking, receipt scanning, and smart donor matching. This project introduces advanced software solutions into a field that has been slow in terms of adopting digital transformations. By reducing the administrative burden and providing deeper insights, this SMS can enable a fairer scholarship distribution. This paper demonstrates how the SMS system enhances efficiency, transparency, and fairness through integrated web and system technologies.

Literature Review

Existing research emphasizes the importance of integrating digital systems into scholarship management due to the expanding volume of applications, the complexity of eligibility criteria, and the demand for transparency. A study by Nguyen-Hoang et al., 2023 proposes a theoretical framework of a scholarship management system that introduces blockchain usage for security and transparency, zero knowledge proofs and self- sovereign identity to validate student identities without sensitive information, and some machine learning. Another article by Harini et al., 2024 highlights the importance of digital transformation in educational management through the implementation of Information and Communication technologies ICM. It discusses different systems such as Learning Management Systems, School Information Management Systems, and financial management applications that enhance administrative tasks, decision making, and educational experience. The authors also shed light on the importance of adequate infrastructure, data protection policies, and staff training. Moreover, a recent study by N.H Ahmad et al., 2024 underscores the importance of efficient, responsive digital platforms for managing financial resources such as scholarships, suggesting that well-designed systems with clear communication and tracking features enhance overall user experience and engagement within online banking and in the context of scholarship management.

Gaps

The research and work done in the field of digitalized scholarship management is significant, yet it presents some gaps. For example, while Blockchain secures data, it also has substantial costs and scalability limitations, especially in high transaction systems. In addition, the use of machine learning is limited to donor- applicant matching and student retention needs. The proposed system in the article provides user specific dashboards but is limited when it comes to

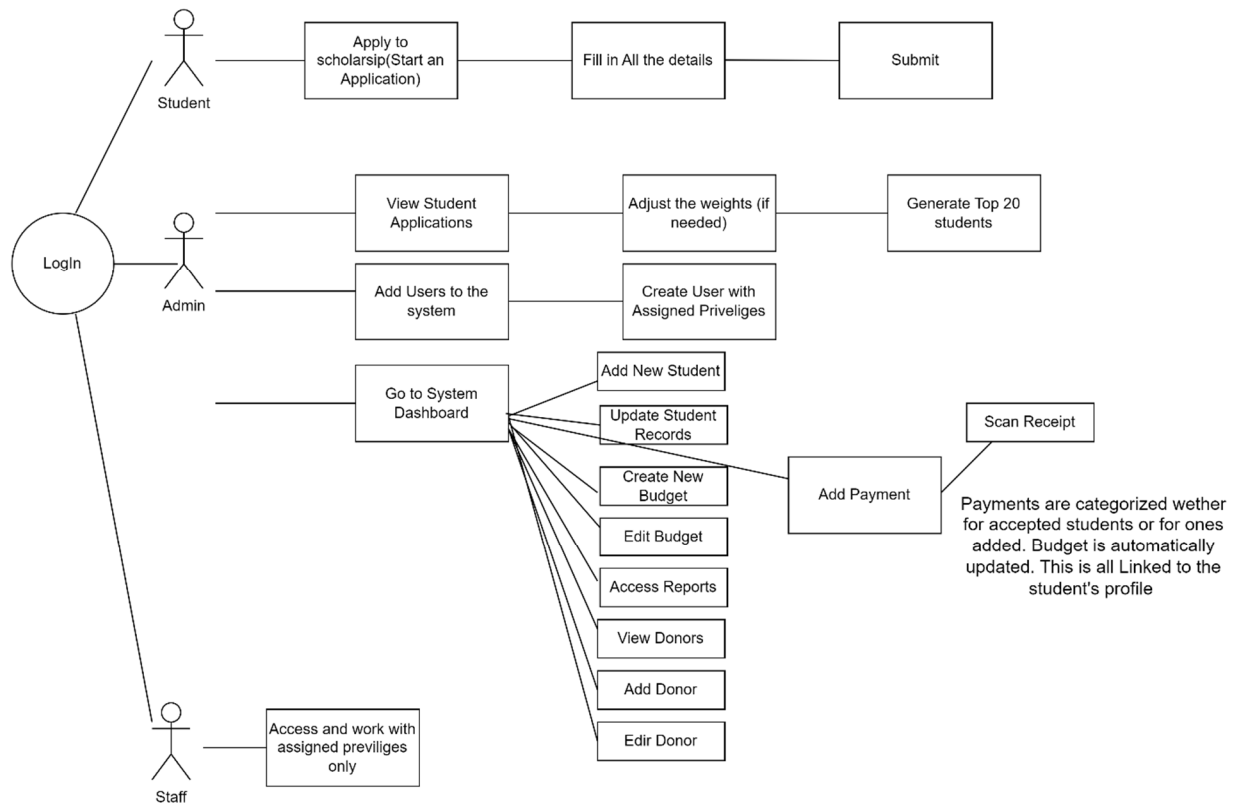
personalized interactive features. Moreover, there is no specific mechanism in any literature yet for budget tracking and allocation or for top eligible students' calculation, and reporting is limited when it comes to system wide analytics and trends. Besides, there is a lack of tools for qualitative analysis of student essays, missing out valuable insights into applicant intentions and needs. In fact, many current platforms lack accessible, engaging user interfaces and fail to offer responsive, mobile-friendly access, particularly critical in regions with varied digital infrastructure. Research shows that transparency in scholarship decisions is limited by manual or fragmented digital tools.

How does the Scholarship Management System fill these gaps?

The smart scholarship management system proposed will not only fill the gaps mentioned but also introduce new features into the digital education systems, especially in Lebanon, where such digital advancements are either fragmented or absent. The SMS system will be accessible across various devices, including mobile phones, laptops, and desktops. It will have a scalable secure cloud-based backend, implementing several features such as dynamic donor matching between donor's preferences and students that match, automated budget allocation and tracking, personalized dashboards for admins, users, and students, advanced analytics and custom reporting, sentiment analysis on application essays, receipt scanning, and most importantly, calculating the top 20 students that are most eligible for scholarships according to several features such as (GPA, Financial need, essay sentiment weight, donor matching weight, employment and dependents weight) and weights that are adjustable by admins according to company's preferences. By integrating these advanced features, that focus the most on transparency, the SMS will introduce a holistic, data-driven approach to scholarship management in Lebanon.

Methodology

The following flowchart represents the main activities of the system and the flow of actions starting as a student, administrator, or a regular assigned member.



The SMS system is a web application that is divided into modules (a modular design) where each function independently yet cohesively contributes to the success of the complex SMS, these modules include:

- Frontend modules: that handle user interfaces (including dashboards, forms, reports, etc.)
- Backend modules: APIs or managements like user authentication, budget, donors, results, reports, nominated candidates, etc.
- Database modules: representing the relational tables done.

Each module handles one responsibility, making the system easier to manage and scale. New modules may be added, or existing ones may be updated without stopping any existing functionality. In addition, the system is easier to update or change since a change in one module does not affect the others. In terms of security, which is very crucial in the system, modules can ensure that such functions are handled in specific isolated modules without being exposed to the other parts of the system.

Design Pattern

The design pattern chosen for the Scholarship management system is the Model View Controller. It presents a clear separation of the system into three interconnected components, it makes the system easier to develop, maintain, and scale. The SMS will be structured using the MVC model in the following way:

1. **Model:** it is responsible for managing the application's data, logic, and rules. It interacts directly with the database. It includes all the database tables needed for the system, and the interactions with these tables (CRUD (Create, Read, Update, & Delete)) operations. In addition, it processes the data by algorithms used for budget tracking, donor matching, and report generation for example, and by certain rules and validations such as the admin has all the privileges and can assign users for the system and grant them access. Moreover, integrating the machine learning part for sentiment analysis on scholarships essays and top 20 recommended individuals will also take place in the model component of the system.
2. **View:** this part handles the presentation layer, the UI. It is responsible for displaying data to users and taking their input. It includes the webpages made with the CSS styling related to them (the dashboards and the privileges). It displays some information fetched

from the database to these webpages. In addition, it handles the add and update forms found in some privileges (students, donors, budgets, etc.). The view part of the system will also ensure a responsive and user-friendly design.

3. **Controller:** The controller component is the connection between the model and the view. It processes the user input, interacts with the model part, and therefore selects the appropriate view to display to the users. It handles the HTTP requests (for API calls or form submissions, etc.). It is also crucial because it validates the form's input before interacting with the database and implements successfully the business logic involved such as ("Only admins can add users") by having session and APIs management.

Example:

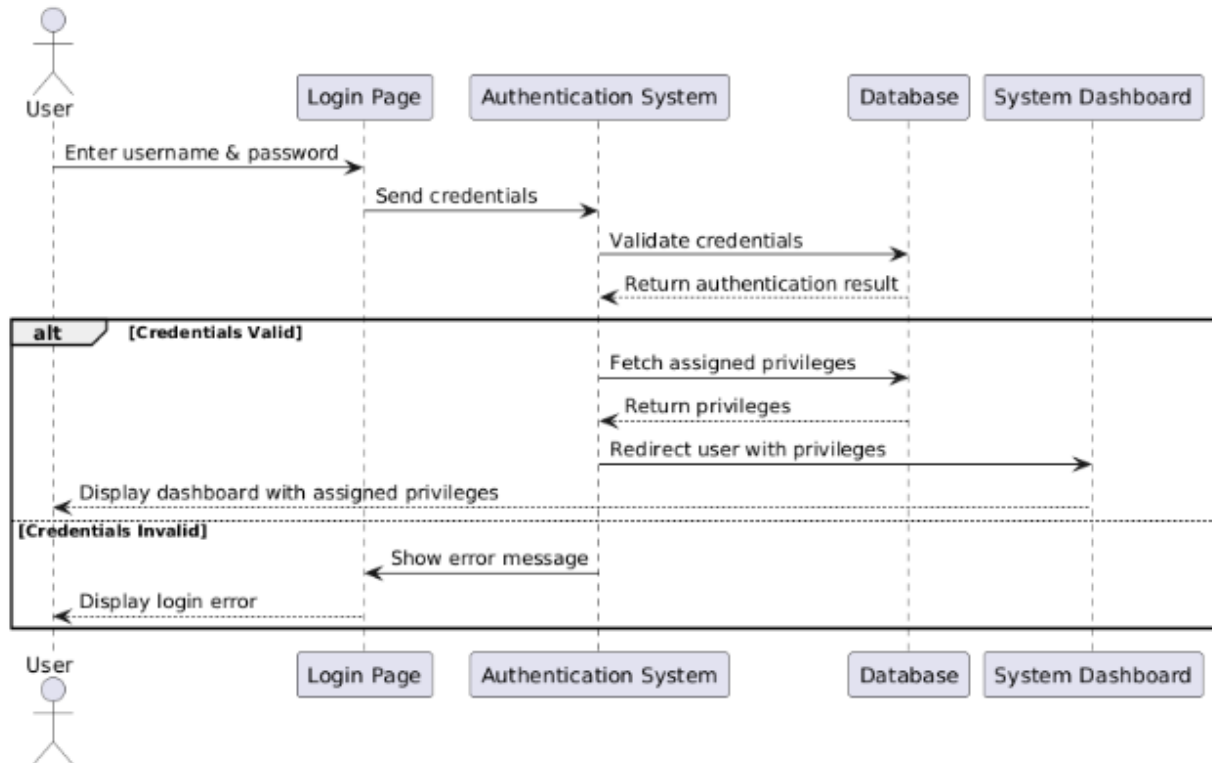
- The Log in page: The **Model** validates the credentials in database, the **View** presents the form for logging in, and the **Controller** authenticates and starts the session based on whether the user now is an admin or an assigned staff member.
- The Add Student feature found in privileges of the system dashboard: The **Model** saves student details in database, the **View** presents the form for adding students, and the **Controller** validates the input and saves it into the database found on PhpMyAdmin.

It is important to note that there are no external or already implemented systems interacting with the SMS other than a hardware piece: a printer.

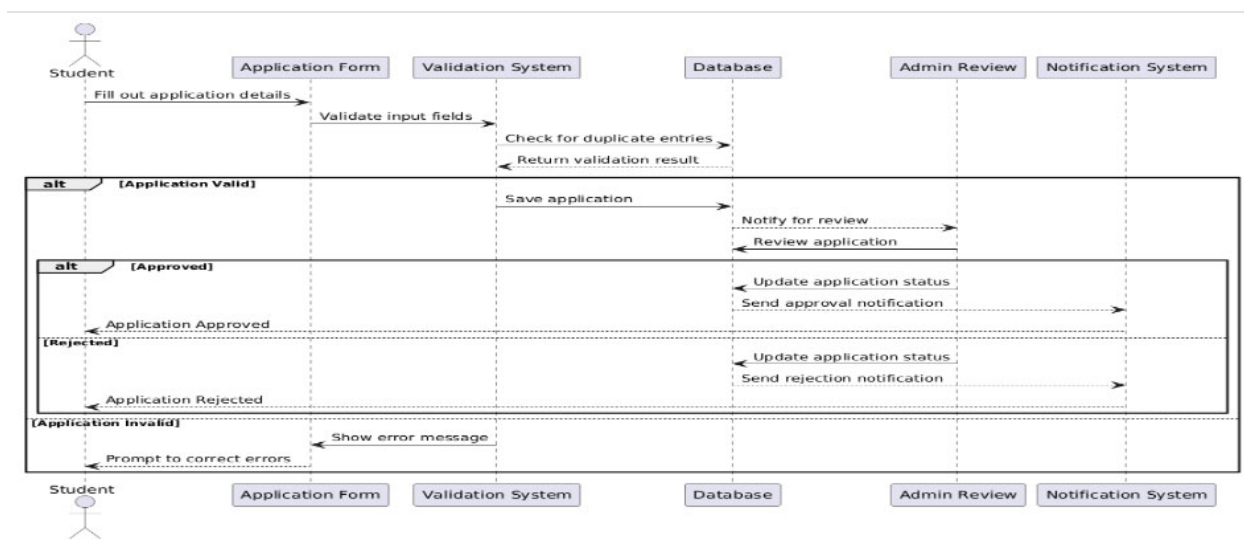
The following UML diagram represents the main database tables and the connections between them.

The following sequence diagram illustrates the process of logging in. The admin logs in with his credentials and can create users and give them access to certain privileges, then upon successful logging they are directed to the system dashboard with the assigned privileges only. The authentication represents the backend logic.

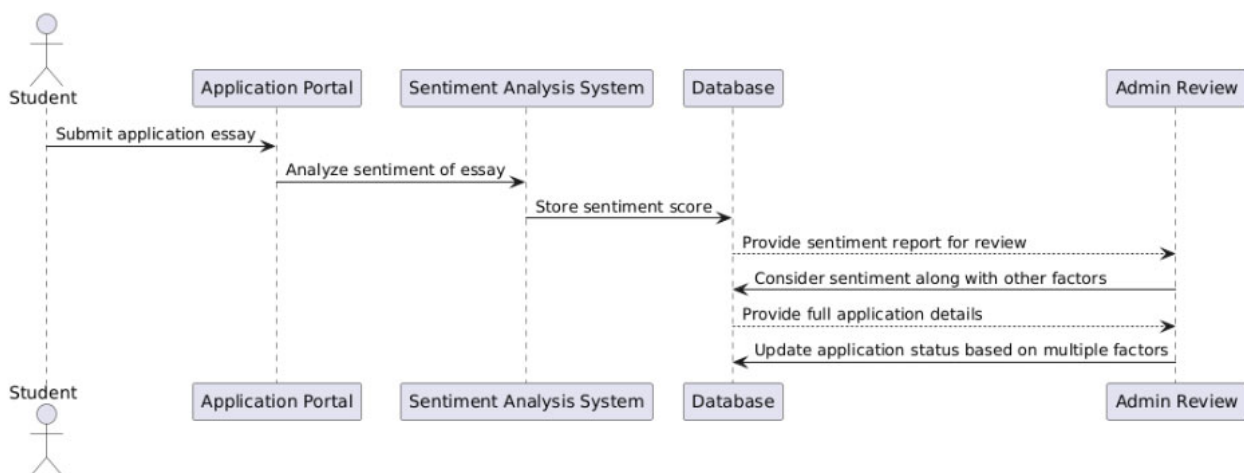
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The following sequence diagram represents the process of students applying for the scholarship. The application is submitted, validated, saved, reviewed, and given a decision. The validation is the logic for syntax and other errors like a duplicate error. The admin has the privilege to access and review the applications (and is helped later with sentiment analysis and other factors that contribute to making a fair decision).

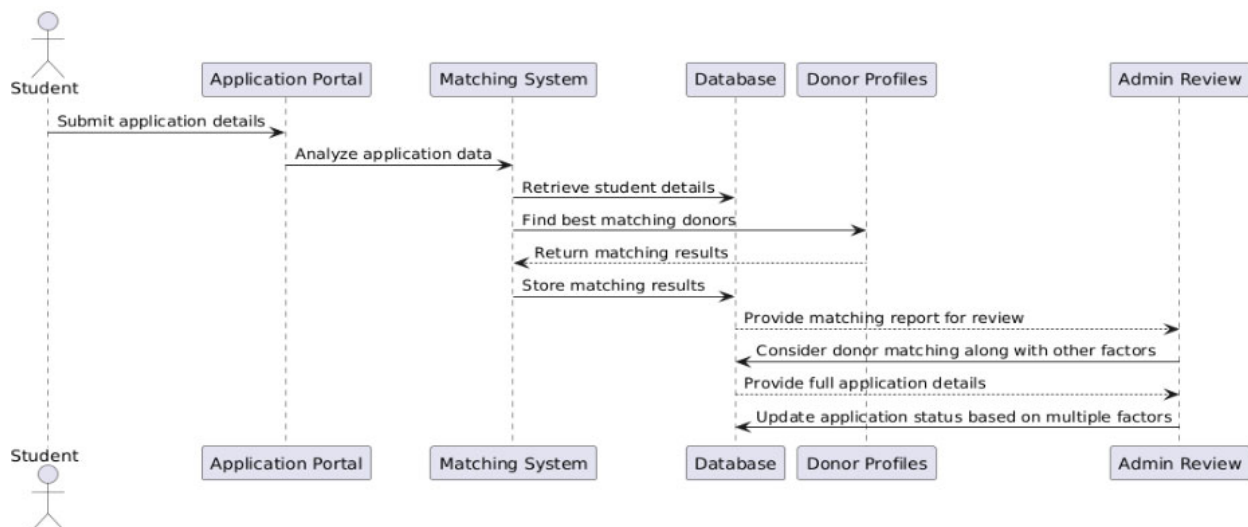


The following sequential diagram represents the process of sentiment analysis on application essays which is a factor that will help in determining the best decision possible on applications statuses. In addition, the sentiment analysis extracts and shows the important words in the essay that lead to a certain score, aiming to achieve the highest transparency possible on every number or step done.



The following sequence diagram represents the process of smart donor matching where specific donors that have preferences similar to some applicants (ready to donate for a specific major for

example) are matched together. The matching system is the backend logic for the data. Donor matching gives a score and is assigned a weight which is a part of the top 20 students calculation.



Security Considerations

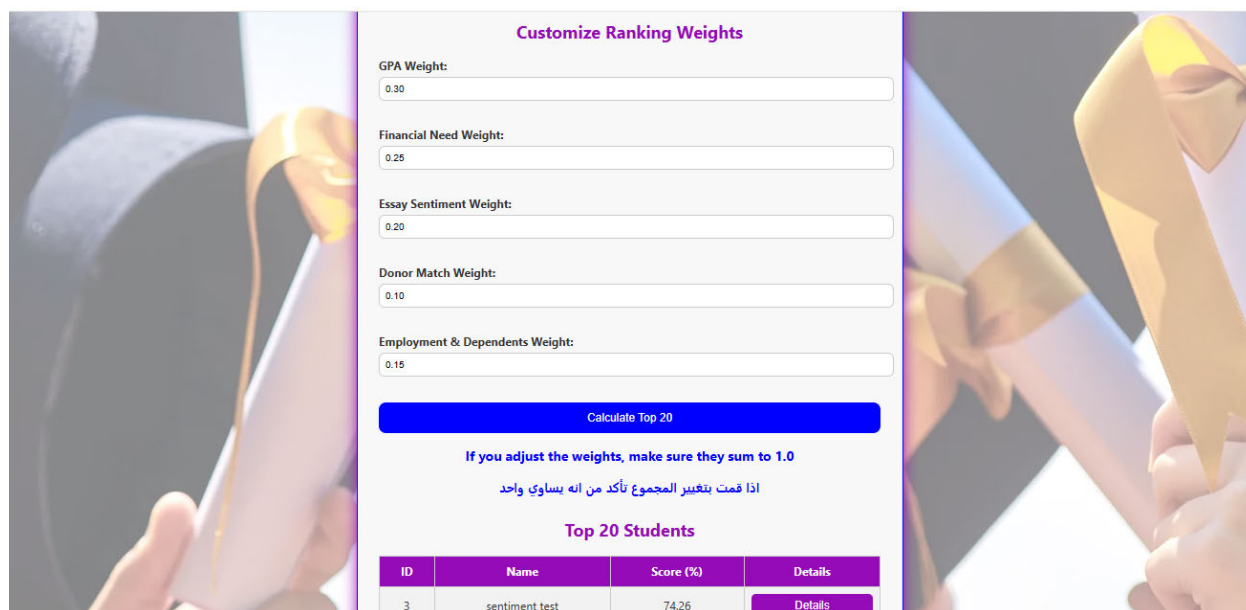
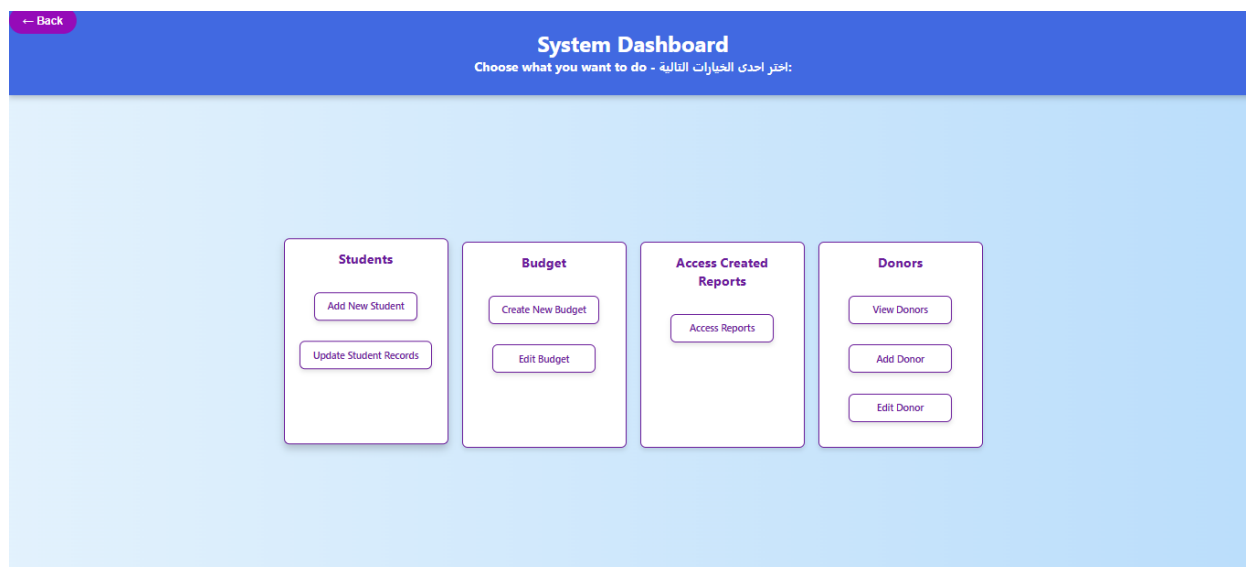
The SMS incorporates several security measures to ensure data integrity, confidentiality, and secure access. First, Role-Based Access Control (RBAC) is implemented to restrict actions based on assigned user roles, ensuring that only authorized individuals can access or modify sensitive information. Encryption is also applied to protect sensitive data such as passwords. Input validation and sanitization prevents SQL injection, cross-site scripting (XSS), and other attacks by ensuring only properly formatted data is accepted. In addition, the system will be hosted online using a private cloud server from AWS, preventing the database from being accessed online, and thus adding more security measures. Regular security audits and system updates that will be made will help identify vulnerabilities and strengthen defenses. These measures collectively enhance the system's security, ensuring the protection of user data and preventing potential breaches.

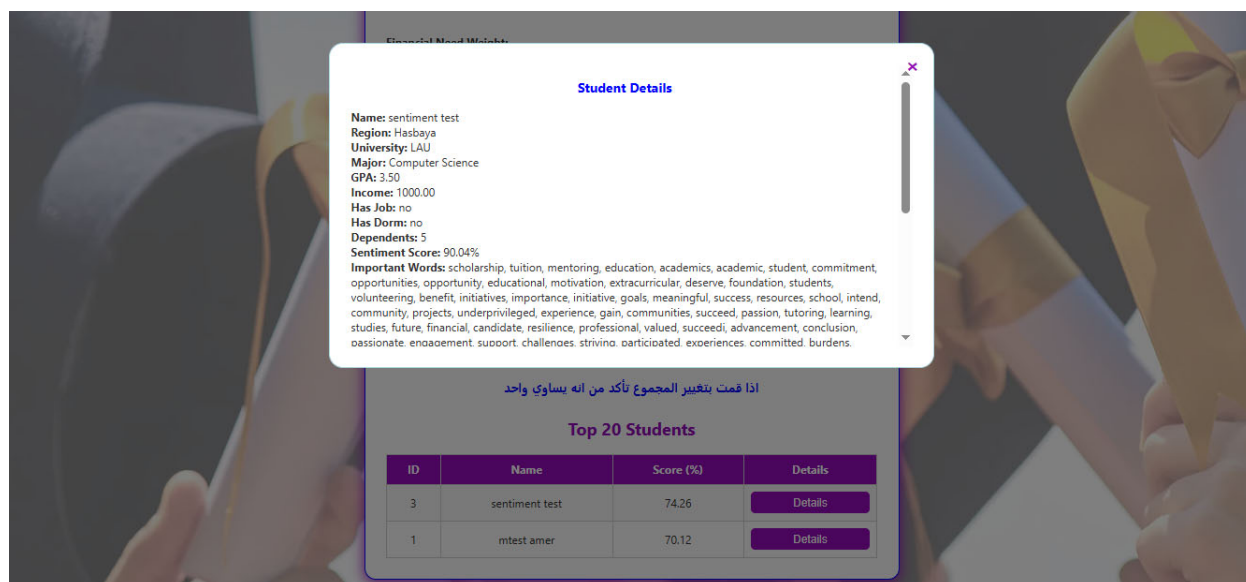
Results and Discussions

The following pictures represent the usability and functional testing of the main components of the system, they show results and functionalities of the system. A discussion part follows.

The screenshot shows two login forms overlaid on a background image of a stack of books and a graduation cap. The top form is titled "Admin / User Login" and contains fields for "Username:" (with "admin" entered) and "Password:". Below these fields is a blue "Login" button. A note below the button states: "If you are a first-time user, ask the admin to create your account!". The bottom form is titled "Student Login" and contains fields for "Email:" and "Password:". A purple information icon is visible on the right side of the background image. At the bottom of the page, a blue footer bar contains the text "© 2025 Scholarship Management System".

The screenshot shows the "Admin Dashboard - الإدارة" interface. It features two main management sections: "Manage Users" and "Manage Scholarships". The "Manage Users" section includes buttons for "Add User" and "Next to Dashboard". The "Manage Scholarships" section includes buttons for "Review Applications" and "Top Candidates". A "Back To Login Page" button is located below these sections. At the bottom of the dashboard, the motivational phrase "Be the Hand that Lifts Others Up!" is displayed in purple text. The background of the dashboard is the same stack of books and graduation cap image seen in the previous screenshot.





The implementation of the Smart Scholarship Management System (SMS) demonstrates a significant advancement in how scholarship decisions are managed and executed. One of the most notable contributions of the system is its ability to introduce transparency, efficiency, and fairness into an area that has historically been subject to human bias, manual errors, and lack of oversight. By leveraging automated scoring mechanisms, sentiment analysis, assessing financial needs, and extracting relevant information from the applications data, the system objectively ranks students and generates a top 20 shortlist based on quantifiable detailed metrics such as GPA, income level, number of dependents, and essay sentiment, which are highly relevant for evaluation. This data-driven approach ensures that scholarship distribution is based on merit and need rather than subjective judgment or favoritism.

The inclusion of sentiment scores and important keywords not only strengthens the evaluation criteria but also reflects the applicant's motivation and alignment with scholarship values and ensures even more transparency showing the details behind the calculated score. This layer of analysis adds qualitative depth to quantitative metrics, offering a holistic view of each candidate.

Furthermore, the system provides immediate access to individual student profiles and highlights key data such as region, job status, dormitory status, and financial background, aiding administrators in making well-informed decisions in a fraction of the time traditional methods would require. Additionally, the system is designed with flexibility that allows organizations to customize the weightings of evaluation criteria according to their unique values and funding priorities. This adaptability ensures the system remains relevant and aligned with diverse organizational missions and evolving scholarship policies.

The system's dashboard architecture further enhances operational efficiency and secure access control. The admin dashboard offers centralized oversight, allowing administrators to manage user accounts, assign privileges, and monitor system-wide activities with ease. Role-based access ensures that users only interact with the features relevant to their responsibilities. For instance, administrators can add or inactivate users, assign specific privileges such as adding or editing student profiles, managing budgets, or generating reports, while staff members can be restricted to their corresponding specific actions. This structured permission system supports data integrity, prevents unauthorized modifications, and facilitates accountability across different user roles. Moreover, the intuitive user interface of the system dashboard allows for seamless navigation and real-time access to critical data, reinforcing the system's commitment to transparency and efficient decision-making.

A particularly valuable feature is the receipt scanning and documentation tool, which allows uploaded receipts to be tied to each student's profile. This functionality adds an essential level of accountability and donor trust. Donors can directly view how funds are being spent, ensuring that their contributions are making a tangible impact. This transparency not only encourages

continued donor engagement but also sets a precedent for ethical financial tracking in nonprofit and educational aid programs.

Overall, the Smart Scholarship Management System addresses critical gaps in traditional scholarship management. Its use of automation, intelligence, and a centralized database transforms the process into a highly efficient, unbiased, and transparent one. As a model, it holds significance in regions such as Lebanon, where NGOs and scholarship granting organizations often lack access to such digital infrastructure. These problems are also present worldwide. The system paves the way for equitable resource distribution and empowers organizations to build long-term trust with both students and donors.

Algorithm for predicting the top 20 students

To predict the top 20 students, a weighted scoring algorithm that evaluates students based on multiple academic and non-academic criteria is implemented. Each criterion is assigned a customizable weight, reflecting its importance in the overall ranking. The formula calculates a composite score for each student by multiplying each attribute by its corresponding weight and summing the results. These scores are then sorted in descending order to identify the top-performing students. The use of adjustable weights provides flexibility, allowing the system to adapt to different institutional priorities or selection criteria, such as placing more emphasis on GPA, extracurricular involvement, or financial need. They might change the weights that sum to 1 and generate a new list of top 20 students with the details of each score shown.

INPUT:

students_list $\leftarrow$ list of students (applicants)

weights $\leftarrow$ weights for each attribute (e.g., GPA, GPA weight = 0.3)

FUNCTION calculate_score(student, weights):

score $\leftarrow$ 0

FOR each attribute, weight IN weights:

 score $\leftarrow$ score + (student[attribute] $\times$ weight)

RETURN score

FOR each student IN students_list:

 student.score $\leftarrow$ calculate_score(student, weights)

SORT students_list BY student.score IN descending order

top_20_students $\leftarrow$ students_list[0 to 19]

OUTPUT: top_20_students

In the implementation of the student ranking algorithm, the default weights, before the possibility of the organization changing them, were assigned to provide the best evaluation criteria based on their relevance to equitable and scholarship decision-making. The weights are as follows: GPA (0.30), financial need (0.25), essay sentiment (0.20), donor match (0.10), and employment and number of dependents (0.15). GPA remains a critical indicator of academic performance and future potential, warranting the highest weight (0.30) as it reflects a student's ability to succeed in an academic environment (Geiser & Santelices, 2007). Financial need is

weighted at 0.25 to ensure that students from underprivileged backgrounds are prioritized, aligning with research that supports need-based allocation to promote access and fairness (Perna, 2006). Essay sentiment, assessed using sentiment analysis, is weighted at 0.20 because it provides insight into the student's motivation, aspirations, and communication skills, which are increasingly used in scholarship evaluations (Nguyen et al., 2017). The donor match is considered at 0.10 to reflect alignment between student goals and donor intentions, which is often a funding prerequisite. Lastly, employment status and number of dependents receive a combined weight of 0.15 to capture socioeconomic challenges and responsibilities outside of academics. These default weights are grounded in both empirical evidence and best practices for scholarship evaluation, offering a balanced framework that values merit, need, and alignment with funding goals.

Conclusion

The Smart Scholarship Management System (SMS) provides an all-in-one solution that is comprehensive, transparent, and equitable platform for managing scholarships. By integrating automated data-driven decision-making, weighted evaluation criteria, several functionalities, role-based access, and a robust backend, the system significantly improves the accuracy, speed, and fairness of the scholarship selection process. Features such as receipt attachment, donor management, and a customizable scoring algorithm encourage trust and accountability, both for administrators and donors. Ultimately, SMS addresses key challenges faced by NGOs and scholarship-granting institutions, particularly in Lebanon, by delivering a scalable, data-driven, and ethical solution. Future enhancements could include the development of a mobile application for greater accessibility, integration with national education databases for real-time data validation, and the implementation of predictive analytics to forecast student success and optimize fund allocation. Additionally, incorporating an AI-based document verification could further expand the system's reach and effectiveness. These improvements would help ensure the system continues to evolve alongside the growing needs of scholarship organizations and student populations. With its well-defined architecture, the system will streamline and revolutionize scholarship management and allocation processes, optimize donor-student matching, and enhance transparency in fund distribution, ultimately improving the efficiency of scholarship management.

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